**TP Report: Generative Models (VAE vs GAN)**

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**1. Introduction**

The objective of this lab is to explore and compare two fundamental families of generative models: Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) . Using the Fashion-MNIST dataset , I implemented both architectures to analyse their trade-offs regarding image quality, training stability, and latent space structure .

**2. Variational Autoencoder (VAE)**

**2.1 Architecture**

I implemented a **Convolutional VAE** as recommended for better feature extraction.

- **Encoder:** Consists of two Conv2d layers (stride 2) followed by ReLU activations. It flattens the output to map the input image (28 x 28) to a probabilistic latent space.
- **Latent Space:** I utilized a dimensionality of **20**. The encoder outputs two vectors: Mean ($\mu$) and Log-Variance ($\log \sigma^2$). I applied the **Reparameterization Trick** ($z = \mu + \sigma \cdot \varepsilon$) to allow backpropagation.
- **Decoder:** Uses ConvTranspose2d layers to upsample the latent vector z back to the original image dimensions. The final activation is Sigmoid, forcing the output pixels into the [0, 1] range.

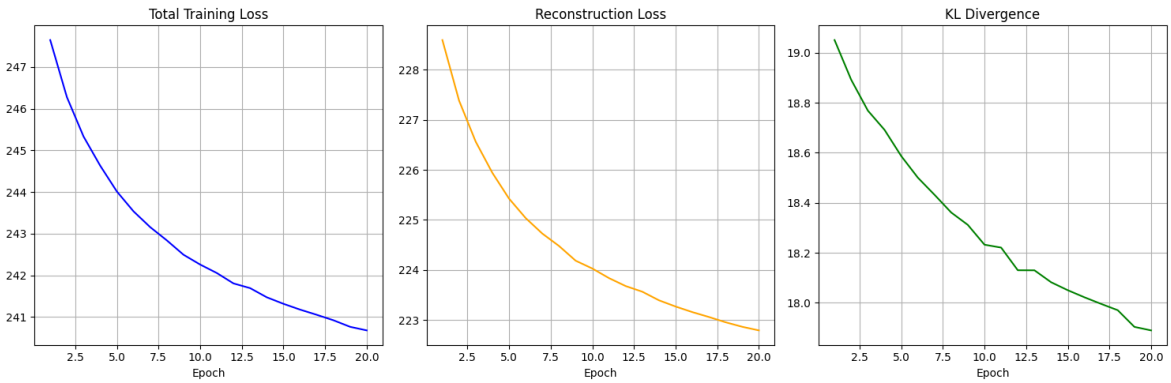
**2.2 Training & Loss Function**

The model was trained for 20 epochs using the **ELBO (Evidence LoIr Bound)** loss function:

1. **Reconstruction Loss:** Binary Cross-Entropy (BCE) to measure pixel-wise similarity.
2. **KL Divergence:** Regularizes the latent space to approximate a Standard Normal Distribution N(0, 1).

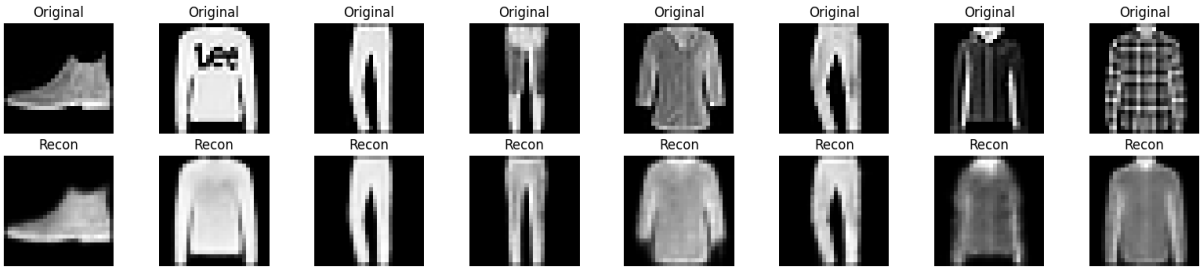
**Training Metrics:**

- **Final Total Loss:** ~240.7
- **Reconstruction Loss:** 218.36 (Indicates successful shape learning).
- **KL Divergence:** 17.81 (Indicates no posterior collapse; the latent space is Ill-structured).

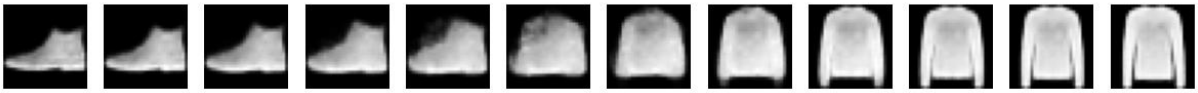


### 2.3 Results

- **Reconstruction:** The model successfully reconstructs the global shape of items (boots, shirts) but struggles with high-frequency details. Texture patterns (e.g., plaid shirts) appear blurred.
- **Interpolation:** Linearly interpolating betIen two latent vectors (e.g., Boot to Shirt) resulted in a smooth semantic transformation. This confirms the latent space is continuous and semantically organized.




Latent Space Interpolation (VAE)



## 3. Generative Adversarial Network (GAN)

### 3.1 Architecture

I implemented a **DCGAN (Deep Convolutional GAN)** architecture.

- **Generator (G):** Takes a noise vector ($z \in \mathbb{R}^{100}$) and upsamples it using ConvTranspose2d layers with Batch Normalization and ReLU. The output activation is **Tanh**, producing images in the range [-1, 1].
- **Discriminator (D):** A binary classifier using strided Conv2d layers and **LeakyReLU (0.2)** to prevent vanishing gradients. It outputs the probability that an image is real.

### 3.2 Stability Improvement

Standard GAN training is unstable. To mitigate this, I implemented **Label Smoothing**.

- Instead of using a hard label of 1.0 for real images, I used **0.9**. This prevents the Discriminator from becoming over-confident, which would otherwise saturate the gradients and stop the Generator from learning.

### 3.3 Training & Results

The model was trained for 20 epochs using the Minimax Loss (Binary Cross-Entropy).

- **Visual Quality:** The GAN produced significantly sharper images than the VAE. Edges of trousers and sneakers are crisp and high-contrast.

- **Diversity:** No severe Mode Collapse was observed; the generator produces a variety of classes (dresses, bags, shoes).
- **Final Metrics:**
    - **D(x) (Real Score):** 0.485 (Target: 0.5)
    - **D(G(z)) (Fake Score):** 0.242
    - *Interpretation:* The Discriminator score near 0.5 for real images indicates a stable equilibrium where the Discriminator is "confused" by the generated samples, proving effective training.

**4. Comparison & Discussion**

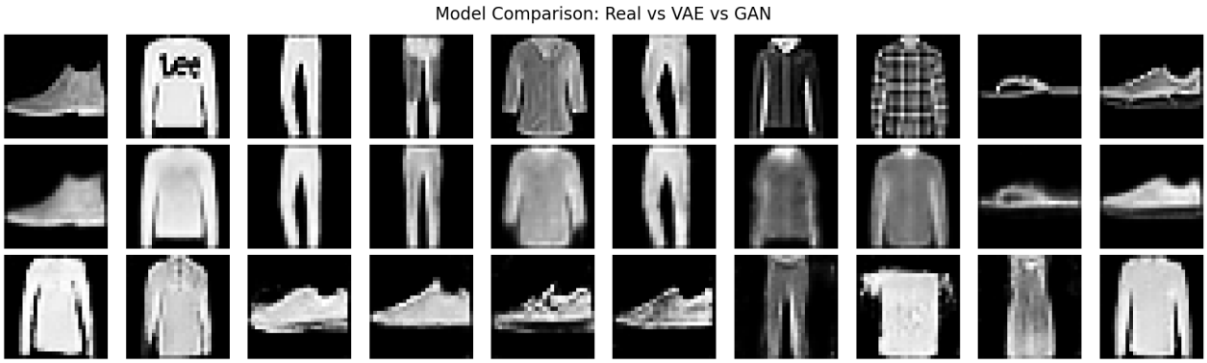
**4.1 Qualitative Comparison**

I compared Real images, VAE reconstructions, and GAN generated samples side-by-side .

| **Feature** | **Variational Autoencoder (VAE)** | **Generative Adversarial Network (GAN)** |
|---|---|---|
| **Image Sharpness** | **Blurry.** Minimizing pixel-wise MSE/BCE leads to averaging, acting as a low-pass filter. | **Sharp.** The adversarial loss penalizes unrealistic blurring, forcing high-frequency details. |
| **Latent Space** | **Continuous.** Smooth interpolation allows morphing betIen objects. | **Discontinuous.** Interpolation often yields nonsensical artifacts. |
| **Training Stability** | **High.** Loss converges monotonically. | **Low.** Loss oscillates; required Label Smoothing to stabilize. |

### 4.2 Limitations

- **VAE:** The primary limitation is the inherent blurriness. It prioritizes covering the entire data distribution (recall) over generating perfectly realistic individual samples (precision).
- **GAN:** While sharper, the GAN occasionally generates geometric artifacts (e.g., disconnected shoe soles) and lacks an explicit inference mechanism (I cannot easily "encode" an image to manipulate it).




### 5. Conclusion

In this lab, I successfully demonstrated the trade-off betIen **density estimation (VAE)** and **sample quality (GAN)**. The VAE proved to be a robust tool for structured representation learning, evidenced by smooth latent interpolations. The DCGAN, stabilized with Label Smoothing, excelled at generating visually convincing, sharp fashion items. Future work could explore **VQ-VAE** or **WGAN-GP** to combine the structural benefits of VAEs with the visual fidelity of GANs.